

HackIT 2.0

Problem statements

1. HealthTech

A. *Title: General Healthcare Service Management in Remote and Rural Areas*

Problem statement:

Healthcare delivery in remote, unreachable and rural areas face significant challenges due to limited medical professionals, infrastructure, huge number of patients etc. Services such as appointments, diagnosis, treatment, follow up, maintaining records done manually leads to delay in certain places is not reachable at remote locations. There is a critical need for a software solution that does patient management, automatic clinical documentation, and monitoring medical services, consultations at timely and effective way that is accessible to all.

B. *Title: IoT-Based Smart Patient Monitoring and Emergency Alert System for Healthcare*

Problem statement:

Healthcare system now a days attempts in ensuring continuous monitoring and timely medical services for patients in hospitals, elderly care centers, and home settings via devices. Sudden health emergencies such as cardiac arrest, oxygen saturation drops, abnormal blood pressure, or accidental falls often go unnoticed due to the absence of real time monitoring and integrated alert mechanisms. Although wearable devices generate health data, many lack centralized dashboards, anomaly detection, emergency response, and secure scalable connectivity. The problem is to design and develop an IoT-based smart patient monitoring and emergency alert system that continuously tracks vital parameters.

C. *Title: Iris Sector Analysis for Disease Prediction Using Colour, Contour, and Pattern Variations*

Problem Statement:

The human iris shows small changes in colour variation, contour, and pattern when certain eye diseases develop. These changes often appear in specific sectors of the iris and may not be noticeable in the early stages. Manual observation by doctors may miss these subtle differences, especially in complex conditions where the symptoms overlap. In addition to analyzing contour distortion and abnormal pattern formation, patient symptoms such as eye pain, blurred vision, redness, and light sensitivity should also be considered. By studying both the sector-wise colour variation and structural changes along with the reported symptoms, the disease can be predicted more accurately and at an earlier stage. Develop a suitable application to implement this.

2. Security

A. *Title: LLM Attacks, LLM Poisoning Detection and Prevention System.*

Problem Statement:

LLMs are now a days used in almost all the user applications (eg: chat bots) and are more prone to vulnerabilities such as prompt injection, data poisoning, malicious fine tuning and adversarial manipulation etc. These lead to misinformation, leakage of sensitive data, wrong outputs and many other issues. Existing security mechanisms that use static filtering and monitoring post deployment are inefficient against these LLM threats. There is a need for reliable system to detect these threats and mitigate them.

B) Title: Anti-Spoofing Face Authentication System for Secure Digital Services

Problem statement:

Face recognition is now popular way for people to authenticate in the applications they use, such as banking apps, workplace platforms, Hostels, e-governance platforms and many more. While it offers ease of use, it also brings security concerns where fraudsters and hackers use printed photos, replayed videos, 3D masks to trick facial authentication systems to get unauthorized access. Therefore we need a system that distinguishes between a real and spoofed attempt to verify genuine users and block fraud attempts in order to maintain user's trust.

3. Smart Automation

A. Title: IoT-Enabled Multi-Robot Warehouse Coordination with Simulation

Problem statement:

Modern warehouses use autonomous robots to transport goods between storage racks and packing areas. As the number of robots increases, problems such as collision, traffic congestion and inefficient routing take place which leads to a heavy delay. A key limitation in these systems is its ability to compute optimal paths dynamically while considering changing task conditions. Industries need efficient, scalable intelligent systems to compute optimal path for a conflict free transportation of goods in the warehouse.

4.FinTech

A. Title: Personalized Financial and Insurance Scheme Recommendation System

Problem statement:

In India, our country financial and insurance needs vary significantly based on individual's age, gender, occupation, income level and also geographic location. Different locations and people residing there (even among themselves), require different schemes. But our government offers a unified scheme to all without any personal assessment. This may lead to lack of services to the people who actually need it, and also excess of gain via those schemes to certain people. Hence develop a data driven system that analyzes personal information and geographic information to identify the correct financial and insurance scheme for individual families.

B. Title: Alternative Credit Scoring for MSMEs

Problem statement:

Micro, Small, and Medium Enterprises (MSMEs) are emerging now a days and play a vital role in India's economy, but still face issues in getting loans etc. Traditional systems approve loan based on past repayment, collateral etc. These might affect potential MSMEs who borrow for first time. A system that checks eligibility based on background verification such as UPI transactions, bank statements, utility payments will help loan lenders to approve for loan. Therefore a data and history driven system to understand their true financial strength and fairly evaluate risk is needed.

5. Student Innovation

A. Title: Society-Oriented Mobile Application using TN MTC Public Data

Problem Statement:

The Tamil Nadu Metropolitan Transport Corporation (TN MTC) provides public datasets related to transportation services. There is an opportunity to develop an innovative mobile application that utilizes this data to address real societal problems, improve public transportation accessibility, and enhance commuter experience.